

Math 141, S10
Exam 3

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	28	
2	20	
3	28	
4	24	
Total	100	

Good Luck !

Problem 1. (28 points) Find each of the following requested limits using l'Hopital's Rule when applicable.

$$(a) \lim_{x \rightarrow 1} \frac{x-1}{x^3-1} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)'}{(x^3-1)'} = \lim_{x \rightarrow 1} \frac{1}{3x^2} = \frac{1}{3}$$

$$(b) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{(e^x - 1 - x)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \rightarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{(e^x - 1)'}{(2x)'} = \lim_{x \rightarrow 0} \frac{e^x}{2}$$

$$= \frac{1}{2}$$

$$(c) \lim_{x \rightarrow +\infty} x e^{-x} \rightarrow \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{x}{e^x} = \lim_{x \rightarrow +\infty} \frac{x'}{(e^x)'} = \lim_{x \rightarrow +\infty} \frac{1}{e^x} = 0$$

$$(d) \lim_{x \rightarrow 0^+} x^{x^2}$$

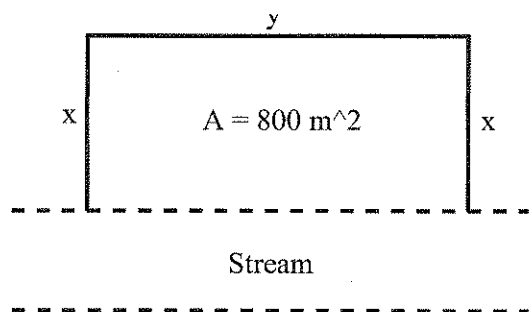
Let $y = x^{x^2}$, then $\ln y = x^2 \ln x$.

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{(\ln x)'}{(\frac{1}{x^2})'}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{(-\frac{2}{x^3})} = \lim_{x \rightarrow 0^+} -\frac{x^2}{2} = 0$$

Hence, $\lim_{x \rightarrow 0^+} x^{x^2} = \lim_{x \rightarrow 0^+} y = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = 1$

Problem 2. (20 points) A rectangle field is to be bounded by a fence on three sides and by a straight stream on the fourth side. Find the dimensions of the field with area $A = 800\text{m}^2$ that uses the shortest length of fencing.



Solution: $L = 2x + y$ $\rightarrow$ length of fencing.

We know $A = 800 \text{ m}^2$, i.e. $xy = 800 \Rightarrow y = \frac{800}{x}$.

Thus

$$L(x) = 2x + \frac{800}{x}, \quad x > 0$$

$$L'(x) = 2 + \left(-\frac{800}{x^2}\right) = 2 - \frac{800}{x^2}$$

Let $L'(x) = 0$, we get

$$2 - \frac{800}{x^2} = 0$$

$$x^2 = 400$$

$$\underline{x = 20\text{m}} \quad (-200 \text{ is impossible since } x > 0)$$

$$\underline{y = \frac{800}{20} = 40\text{m}}$$

Problem 3. (28 points) Evaluate each of the following indefinite/definite integrals.

$$\begin{aligned}
 \text{(a)} \quad & \int (x^7 - 2x^{-3} - \sqrt[3]{x}) dx \\
 &= \frac{x^8}{8} - 2 \cdot \frac{x^{-2}}{(-2)} - \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + C \\
 &= \frac{x^8}{8} + \frac{1}{x^2} - \frac{3}{4} x^{\frac{4}{3}} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & \int 2x(x^2 - 2)^9 dx \\
 & \text{Let } u = x^2 - 2, \text{ then } du = 2x dx. \\
 & \int 2x(x^2 - 2)^9 dx = \int u^9 du = \frac{u^{10}}{10} + C \\
 &= \frac{(x^2 - 2)^{10}}{10} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad & \int \frac{\cos t}{1 + \sin^2 t} dt \\
 & \text{Let } u = \sin t, \text{ then } du = \cos t dt \\
 & \int \frac{\cos t}{1 + \sin^2 t} dt = \int \frac{1}{1 + u^2} du \\
 &= \arctan u + C \\
 &= \arctan(\sin t) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & \int_e^{e^2} \frac{1}{x \ln x} dx \\
 & \text{Let } u = \ln x, \text{ then } du = \frac{1}{x} dx \\
 & x = e, u(e) = \ln e = 1 \\
 & x = e^2, u(e^2) = \ln e^2 = 2 \ln e = 2 \\
 & \int_e^{e^2} \frac{1}{x \ln x} dx = \int_1^2 \frac{1}{u} du = \ln u \Big|_1^2 = \ln 2 - \ln 1 \\
 &= \ln 2 - 0 = \ln 2
 \end{aligned}$$

Problem 4. (24 points)

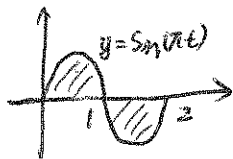
(a) Let $v(t) = \sin(\pi t)$ (measured in feet per second) be the velocity of a particle moving along a coordinate line. Find the displacement of the particle during the time period $0 \leq t \leq 2$ and the distance traveled during this time period.

Solution: The displacement

$$\int_0^2 \sin(\pi t) dt = \left[-\frac{\cos(\pi t)}{\pi} \right]_0^2 = \left(-\frac{\cos(2\pi)}{\pi} \right) - \left(-\frac{\cos(0)}{\pi} \right) = \frac{1}{\pi} - \frac{1}{\pi} = 0 \text{ ft}$$

The distance traveled.

$$\begin{aligned} \int_0^2 |\sin(\pi t)| dt &= \int_0^1 \sin(\pi t) dt + \int_1^2 (-\sin(\pi t)) dt \\ &= \left[-\frac{\cos(\pi t)}{\pi} \right]_0^1 + \left[\frac{\cos(\pi t)}{\pi} \right]_1^2 \\ &= \left[-\frac{\cos \pi}{\pi} + \frac{\cos 0}{\pi} \right] + \left[\frac{\cos 2\pi}{\pi} - \frac{\cos \pi}{\pi} \right] \\ &= \frac{2}{\pi} - \frac{2}{\pi} \\ &= \frac{4}{\pi} \text{ ft} \end{aligned}$$



(b) Find the area A of the region bounded by the curves $y = e^{2x}$ and $y = x^2$, and bounded on the sides by $x = 0$ and $x = 1$.

Solution:

$$\begin{aligned} A &= \int_0^1 (e^{2x} - x^2) dx \\ &= \left[\frac{e^{2x}}{2} - \frac{x^3}{3} \right]_0^1 \\ &= \left[\frac{e^2}{2} - \frac{1}{3} \right] - \left[\frac{e^0}{2} - 0 \right] \\ &= \frac{e^2}{2} - \frac{1}{3} - \frac{1}{2} \\ &= \frac{e^2}{2} - \frac{5}{6} \end{aligned}$$

Extra Credits. (5 points) Evaluate $\int \frac{x}{\sqrt{1-x^4}} dx$.

Solution: Let $u = x^2$, then $du = 2x dx$

$$\Downarrow \\ \frac{1}{2} du = x dx$$

$$\int \frac{x}{\sqrt{1-x^4}} dx = \int \frac{1}{\sqrt{1-u^2}} \left(\frac{1}{2}\right) du$$

$$= \frac{1}{2} \arcsin u + C$$

$$= \frac{1}{2} \arcsin x^2 + C$$